Hide menu

Week 3: Image Sensing

Reading: Week 3 Lecture Handout
30 min

Ungraded Plugin: 3.1 Overview of Imaging Sensing
3 min

Practice Quiz: 3.1 Overview of Image Sensing Self-check Quiz
Started

Ungraded Plugin: 3.2 A Brief History of Imaging
16 min

Practice Quiz: 3.2 A Brief History of Imaging Self-check Quiz
Started

Reading: 3.3 Video Correction
10 min

Ungraded Plugin: 3.3 Types of Image Sensors
11 min

Practice Quiz: 3.3 Types of Image Sensors Self-check Quiz
Started

Reading: 3.4 Video Correction
10 min

Ungraded Plugin: 3.4 Resolution, Noise, Dynamic Range
14 min

Practice Quiz: 3.4 Resolution, Noise and Dynamic Range Self-check Quiz
Started

Ungraded Plugin: 3.5 Sensing Color
19 min

Reading: 3.5 Sensing Color Supplementary Reading
10 min

Practice Quiz: 3.5 Sensing Color Self-check Quiz
Started

Ungraded Plugin: 3.6 Camera Response and HDR Imaging
17 min

Practice Quiz: 3.6 Camera Response and HDR Imaging Self-check Quiz
10 min

Ungraded Plugin: 3.7 Nature's Image Sensors
8 min

Discussion Prompt: Week 3 Designing a Camera Lens
10 min

Discussion Prompt: Week 3 Questions and Feedback

Week 3 Quiz

Congratulations! You passed!

Grade received 100% **To pass** 50% or higher

Go to next item

3.6 Camera Response and HDR Imaging Self-check Quiz

coach

Ready to review what you've learned before taking the quiz? I'm here to help.

Help me practice Let's chat

Submit your assignment

Try again

Receive grade

To Pass 50% or higher

Your grade 100%

View Feedback

We keep your highest score

Like Dislike Report an issue

1. The measured brightness is linearly related to incoming photon flux. **1 / 1 point**

- True
- (selected) False

Correct
Cameras introduce some non-linearity during processing.

2. Why would cameras introduce non-linearity to measured brightness? **1 / 1 point**

- To increase the dynamic range
- To make images more or less bright
- (selected) To compress some brightness ranges more than other brightness ranges

Correct
For example, we may want to compress the dark parts differently than the bright parts of the image

3. What is the main issue with enhancing the dynamic range of a scene using multiple images with different exposures? **1 / 1 point**

- The highest brightness value crosses 255
- The images are not clear
- (selected) Does not work well for moving images as it essentially adds up multiple images

Correct
Using multiple images with different exposures works well for static images but poorly for moving images.